ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so τ_0 must be between -2.4525 and +2.4525 N m.

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state $\dot{\theta} = 0$, and I assume $\theta ss = \theta d = 90°$ because the PD controller tracks the desired angle. Thus the steady-state error is zero.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

$$y[k] = 0.8818y[k-1] + 0.0591x_k + 0.0591x[k-1], DCgain1$$

Q3(c) [4 marks]

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so θss=1 and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.